

(Valence-shell electron-pair repulsion theory) VSEPR theory: predicting molecular geometries.

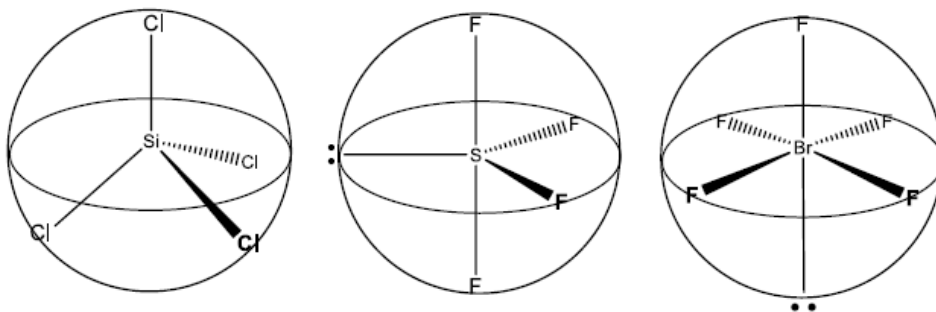
In all chemical combinations there occurs an interaction between outer orbital electrons of the reactants. The electrons belonging to the outermost quantum shell of the central reactant atom plus the additional electrons acquired as a result of bonding are called valence shell electrons. According to VSEPR theory the arrangement of the bonds around an atomic centre depends on the number of electron pairs surrounding the central atom, and also on the relative size and shape of the orbitals which contain valence shell electron pairs. The preferred arrangement of a given number of electron pairs in the valence shell of an atom is the one that maximises their distance apart. **Each electron pair occupies a well-defined region of space and other electrons avoid this space. Note that the linear, triangular planar, regular tetrahedral, trigonal bipyramidal and regular octahedral arrangements of electron pairs with 2, 3, 4, 5 and 6 bonding pairs (STERIC NUMBER).** By making additional postulates it is possible to explain small variations in bond angles and bond lengths from those predicted by theory. It needs to be noted that the electron pairs in valence shell are not all equivalent. They may bond different ligands, they may be unshared lone pairs or they may participate in multiple bonding. Some rules are now given to help explain the small variations in bond angles of certain common covalent molecules.

**Remember
while
calculating
STERIC
NUMBER**

The electrons are arranged in pairs that are either bonding (shared) pairs [X] or non bonding (unshared) pairs [E]. In the usual σ - π treatment this usually means ignoring the π bonds temporarily since they will follow the σ bonds and therefore the two bonding pairs of a double bond or three bonding pairs of a triple bond are all primarily considered as a single point on the sphere.

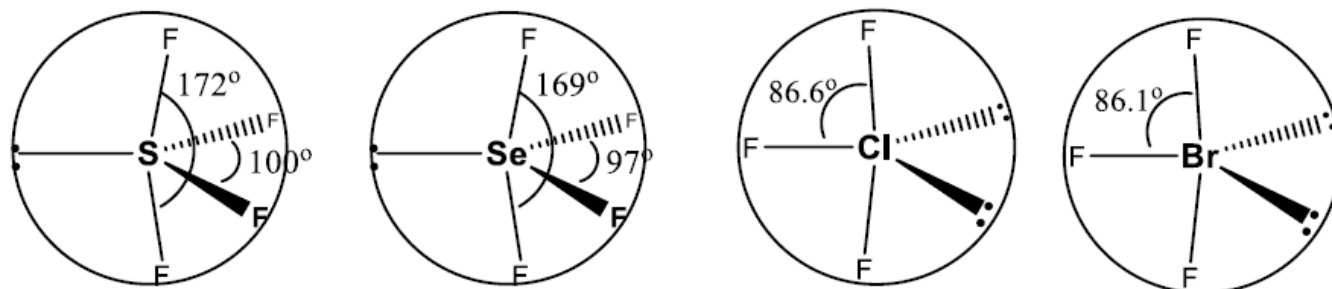
The rules

The pairs of electrons in the valence shell adopt that arrangement which maximizes their distances apart. This in other words means that the bonding and non bonding electron pairs are arranged as far apart from each other as possible on the surface of a sphere with the atom A at the centre of the sphere.

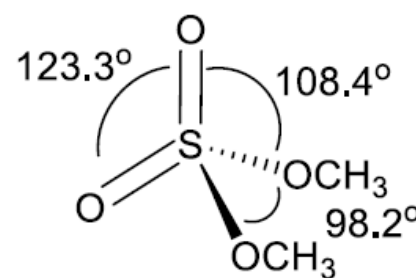
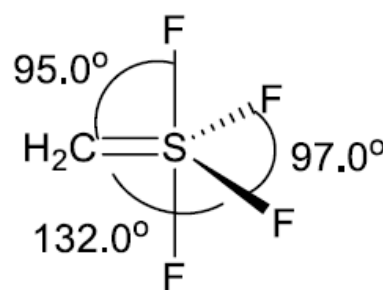
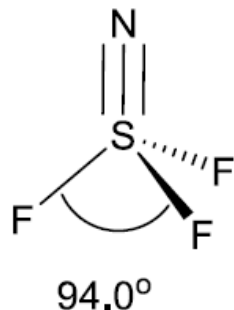
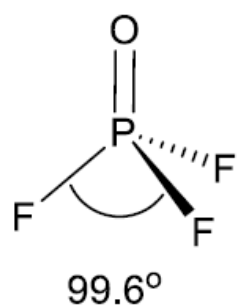


Rules which take care of the deviations observed from the ideal geometry

Rule 1: A non bonding lone pair [E] occupies more space on the surface of the central atom than a bonding pair [X]. In other words, lone pair- lone pair repulsion is greater than lone pair- bond pair repulsion which is greater than bond pair-bond pair repulsion. Lone pairs choose the largest site, e.g., **equatorial in trigonal bipyramid and axial in pentagonal bipyramid**. If all sites are equal, lone pairs will be better placed trans to each other.



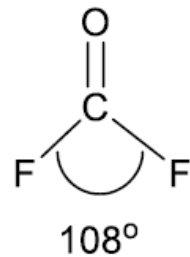
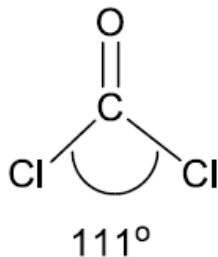
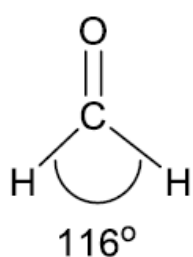
Rule 2: The strengths of the repulsions between bonds of different multiplicity decreases in the order: triple–double > double– double > double–single > single–single.

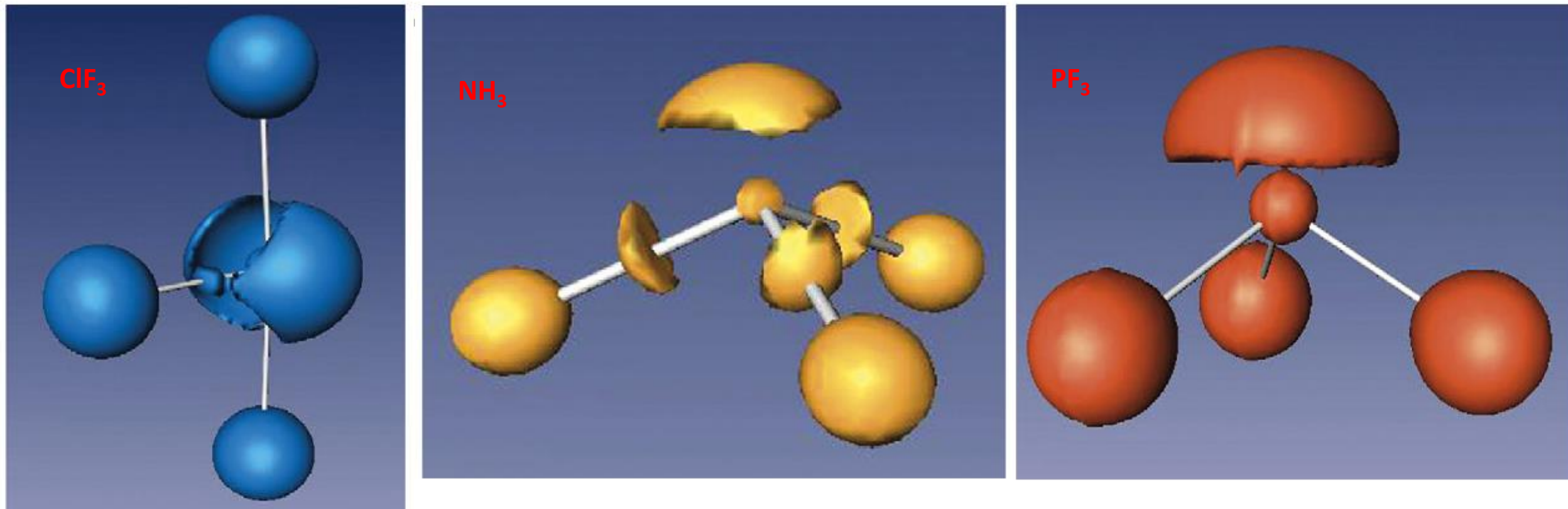


Both $\text{P}(\text{O})\text{F}_3$ and $\text{S}(\text{N})\text{F}_3$ have tetrahedral geometry, however $\text{F}-\text{E}-\text{F}$ ($\text{E} = \text{P}$ or S) are different. Explain?

Rule 3: Bonding pairs to electronegative substituents occupy less space than those to more electropositive substituents. Therefore angles between the single bonds, decrease with increasing electronegativity of the substituent.

This decrease in domain size with increasing electronegativity of the substituent happens as more of the bonding electron density is shifted to the electronegative substituent. In other words, the repulsion between single bonds decreases with increasing electronegativity of the substituent and/or decreasing electronegativity of the central atom.





(a) ClF_3 — showing the charge concentrations corresponding to the three F atoms and the two charge concentrations corresponding to the two lone pairs in the valence shell of the central Cl atom.

(b) NH_3 — showing the nitrogen lone pair, three bonding pairs and the three H atoms.

(c) PF_3 — showing the phosphorus lone pair and the three F atoms – no bonding charge concentrations are observed because of the large difference in electronegativity between P and F.

(1) Predict the shape of HCN molecule

Step1: Carbon is the central element, based on Lewis model predicts the bond between C and H, and also for C and N

Step 2: Check if there is any lone pair and find out the number of sigma bond

Step 3: Predicts the shape of HCN

(2) Justify the tetrahedral shape of $[\text{BH}_4]^-$

Hints: Anion indicate one electron extra to the central atom

(3) Predicts the shapes of chlorate ion (ClO_3^-)

(4) Predicts the shape of PF_5

1. Find the molecule with the least bond angle

- (a) BeF_2
- (b) CH_4
- (c) NH_3
- (d) H_2O

2. Find the species with a maximum bond angle

- (a) NH_4^+
- (b) SCl_2
- (c) NH_3
- (d) PCl_3

3. Which of the following has a linear shape? (JEE)

- (a) NO_2^-
- (b) NO_2^+
- (c) O_3
- (d) SO_2

Practice questions

- (1) The Lewis acidity of BF_3 is lesser than that of BBr_3 . Why?
- (2) $(\text{SiH}_3)_3\text{N}$ is a weaker base while $(\text{CH}_3)_3\text{N}$ is stronger. Why?
- (3) Between NH_3 and NF_3 which will have a higher dipole moment. Which will have a higher bond angle? Justify.
- (4) Solvation and hydration energy increases with increasing ionic potential. Comment (Define ionic potential)
- (5) Isostructural species are those which have the same shape and hybridisation. Among the given species identify the isostructural pairs.
(a) NF_3 and BF_3 (b) $[\text{BF}_4]^-$ and $[\text{NH}_4]^+$ (c) BCl_3 and BrCl_3 (d) None of the above
- (6) $\text{K}_4[\text{Fe}(\text{CN})_6]$ is diamagnetic, however $\text{K}_4[\text{FeF}_6]$ is paramagnetic, explain?

Bonding in d-block metal complexes: valence bond theory

The atomic orbitals required for hybridization in an octahedral complex are the $3d_{z^2}$, $3d_{x^2-y^2}$, $4s$, $4p_x$, $4p_y$ and $4p_z$ (Table 20.1); these orbitals must be unoccupied so as to be available to accept six pairs of electrons from the ligands. The Cr^{3+} ion has three unpaired electrons and these are accommodated in the $3d_{xy}$, $3d_{xz}$ and $3d_{yz}$ orbitals:

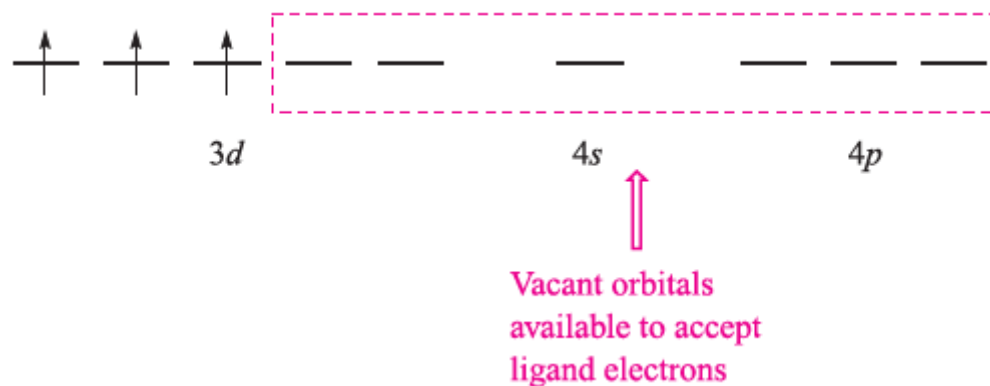
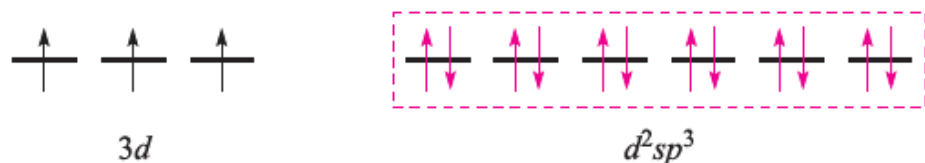


Table 20.1 Hybridization schemes for the σ -bonding frameworks of different geometrical configurations of ligand donor atoms.

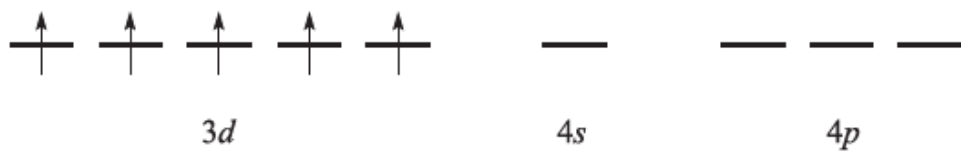
Coordination number	Arrangement of donor atoms	Orbitals hybridized	Hybrid orbital description	Example
2	Linear	s, p_z	sp	$[\text{Ag}(\text{NH}_3)_2]^+$
3	Trigonal planar	s, p_x, p_y	sp^2	$[\text{HgI}_3]^-$
4	Tetrahedral	s, p_x, p_y, p_z	sp^3	$[\text{FeBr}_4]^{2-}$
4	Square planar	$s, p_x, p_y, d_{x^2-y^2}$	$sp^2 d$	$[\text{Ni}(\text{CN})_4]^{2-}$
5	Trigonal bipyramidal	s, p_x, p_y, p_z, d_z^2	$sp^3 d$	$[\text{CuCl}_5]^{3-}$
5	Square-based pyramidal	$s, p_x, p_y, p_z, d_{x^2-y^2}$	$sp^3 d$	$[\text{Ni}(\text{CN})_5]^{3-}$
6	Octahedral	$s, p_x, p_y, p_z, d_z^2, d_{x^2-y^2}$	$sp^3 d^2$	$[\text{Co}(\text{NH}_3)_6]^{3+}$

With the electrons from the ligands included and a hybridization scheme applied for an octahedral complex, the diagram becomes:

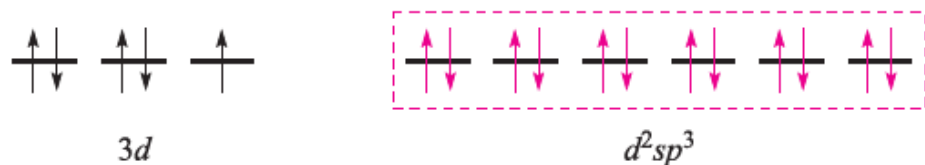


This diagram is appropriate for all octahedral Cr(III) complexes because the three 3d electrons always singly occupy different orbitals.

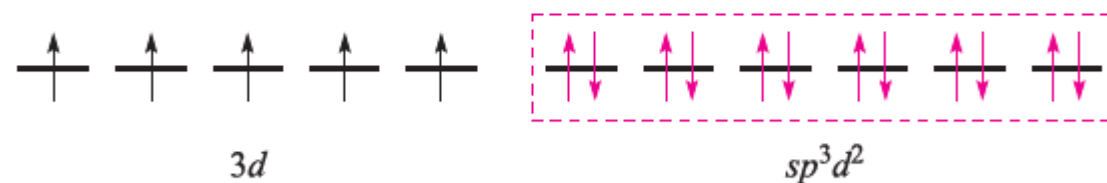
For octahedral Fe(III) complexes, we must account for the existence of both high- and low-spin complexes. The electronic configuration of the free Fe^{3+} ion is:



For a low-spin octahedral complex such as $[\text{Fe}(\text{CN})_6]^{3-}$, we can represent the electronic configuration by means of the following diagram where the electrons shown in red are donated by the ligands:



For a high-spin octahedral complex such as $[\text{FeF}_6]^{3-}$, the five 3d electrons occupy the five 3d atomic orbitals (as in the free ion shown above) and the two d orbitals required for the sp^3d^2 hybridization scheme must come from the 4d set. With the ligand electrons included, valence bond theory describes the bonding as follows leaving three empty 4d atomic orbitals (not shown):



This scheme, however, is unrealistic because the 4d orbitals are at a significantly higher energy than the 3d atomic orbitals.

Valence bond theory may rationalize stereochemical and magnetic properties, but only at a simplistic level. It can say nothing about electronic spectroscopic properties (color of compounds).

Finally, it cannot tell us why certain ligands are associated with the formation of high- (or low-) spin complexes.

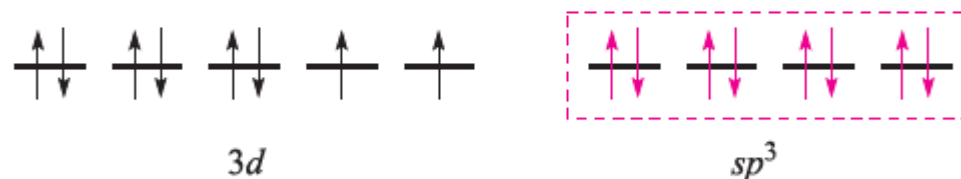
We therefore move on to alternative approaches to the bonding.

Verify that Ni(II) forms paramagnetic tetrahedral and octahedral complexes however the square planer complex of Ni(II) is diamagnetic.

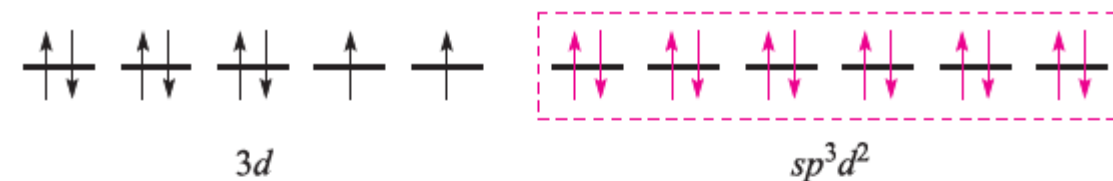
A paramagnetic compounds are those which has unpaired electron

A diamagnetic complex are those which does not have any unpaired electron

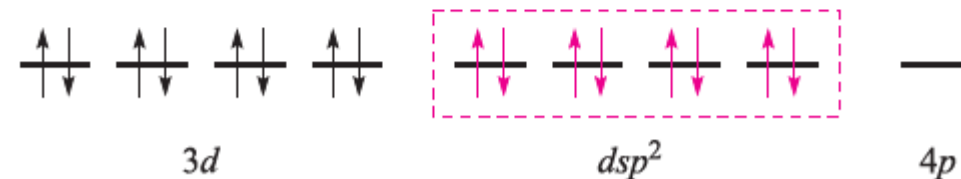
Nickel(II) (d^8) forms paramagnetic tetrahedral and octahedral complexes, and diamagnetic square planar complexes. Bonding in a tetrahedral complex can be represented as follows (ligand electrons are shown in red):



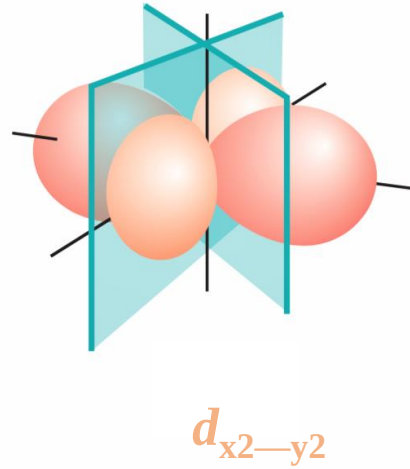
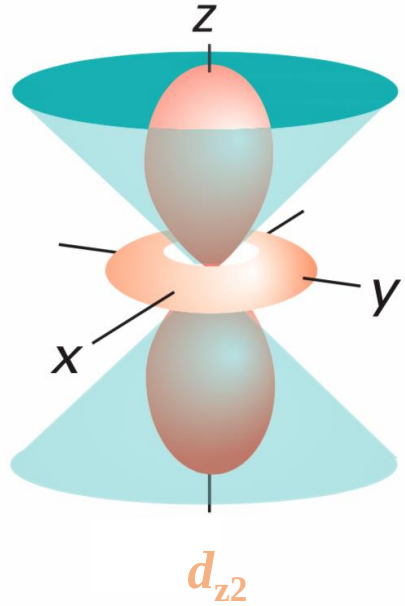
and an octahedral complex can be described by the diagram:



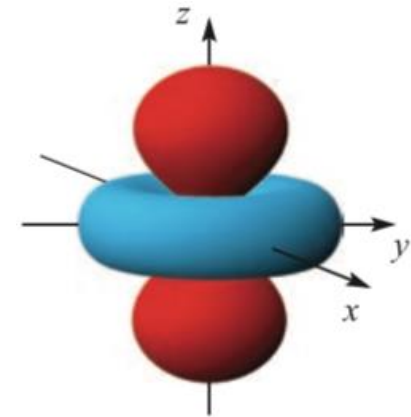
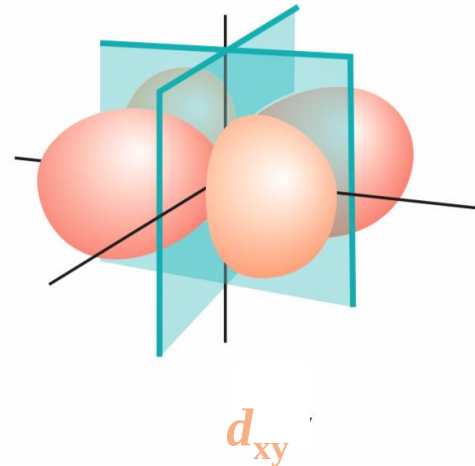
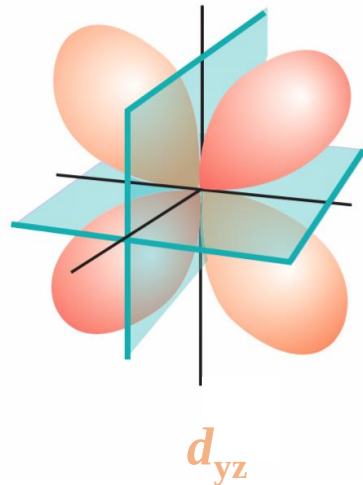
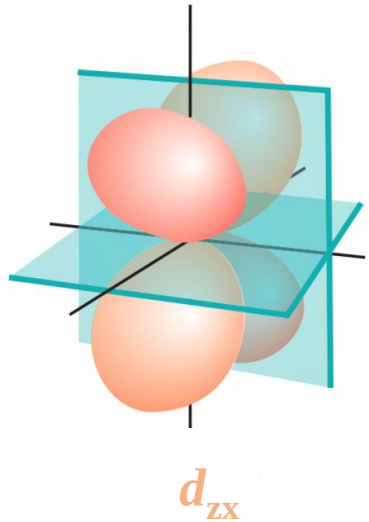
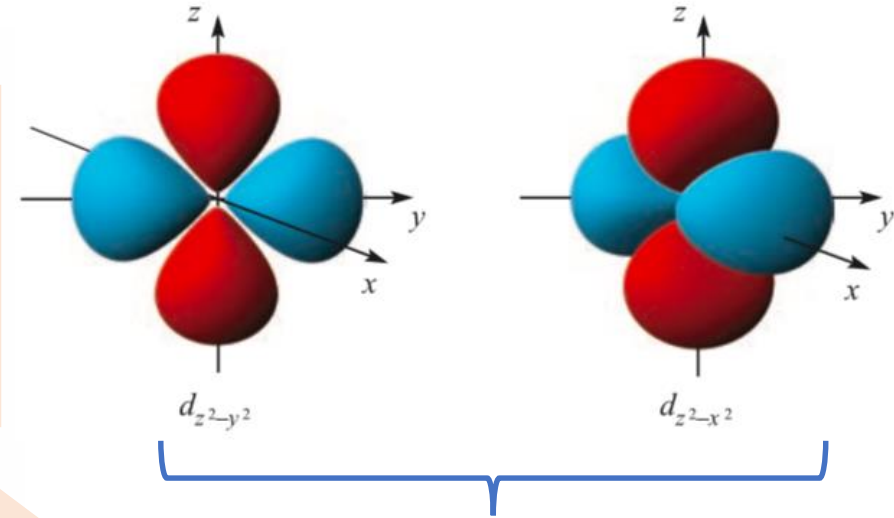
in which the three empty $4d$ atomic orbitals are not shown. For diamagnetic square planar complexes, valence bond theory gives the following picture:



The conventional representation of a set of five degenerate d -orbitals



The formation of a d_{z^2} orbital from a linear combination of $d_{z^2-y^2}$ and $d_{z^2-x^2}$ orbitals.



Theories of bonding in transition metal complexes – in a nutshell

Theories of bonding were proposed and used to explain the observed properties of transition metal complexes such as color, magnetism, shape of complexes

Valence bond theory

- Based on the concept of hybridization sp^3 , dsp^2 , dsp^3 , d^2sp^3 & sp^3d^2 : Predicts shapes of complexes very efficiently
- Can determine magnetic moment if hybridization is known and vice versa
- Does not explain color of complexes, distortion of shape of complexes
- Does not predict strength of ligands or temperature dependence of magnetic moments

Crystal Field theory

- Basic assumptions: Ligands and metal are point charges and the attraction between them is purely electrostatic in nature
- Considers how the energies of the five metal d orbitals change in the presence of a ligand field (removing /lifting of the degeneracy).
- Provides explanation to color, arranges ligands according to their strength, explains distortion of complexes and anomalies in their physical properties
- **VBT and CFT should not be MIXED**: A common mistake done by many students

Molecular Orbital Theory

- A larger picture where both metal orbitals and *ligand group orbitals* are made to form bonding, non bonding and antibonding orbitals.
- CFT splitting is included in this picture; also orbital overlap: π bonding
- Explains color, magnetism and energetics (does not predict shape)



Linus Pauling; Nobel 1954,1962



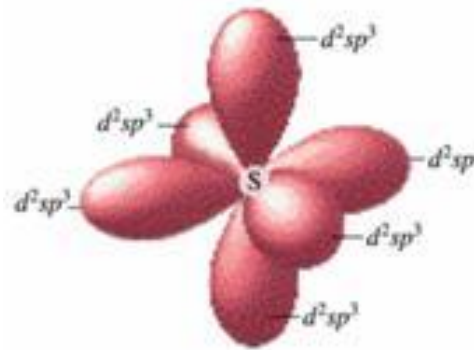
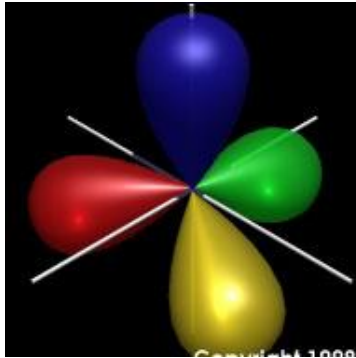
Hans Bethe, Nobel 1967



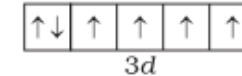
Mulliken & Hund, Nobel 1966(Mulliken)

Theories of bonding in transition metal complexes

Valence bond theory



Orbitals of Co^{3+} ion

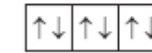


d^2sp^3 hybridised orbitals of Co^{3+}



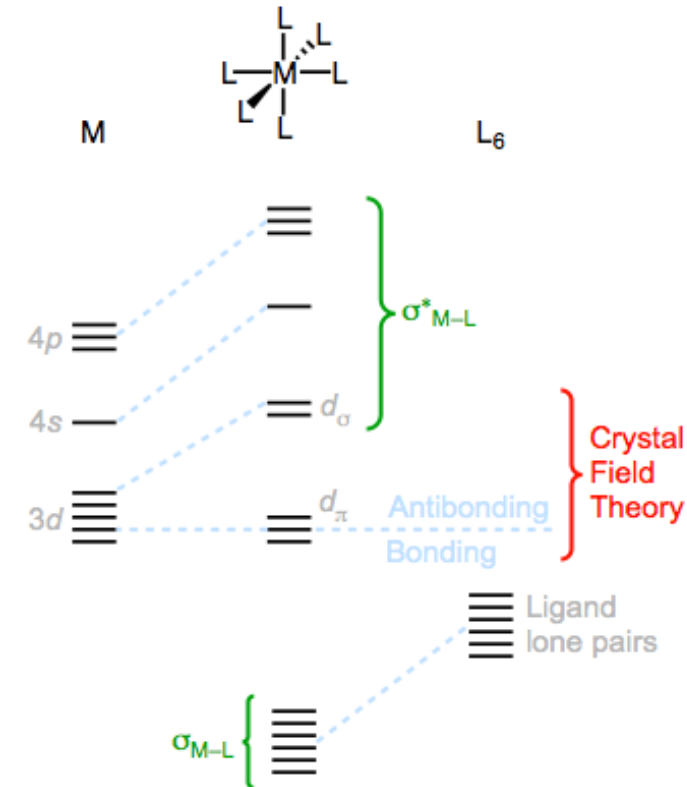
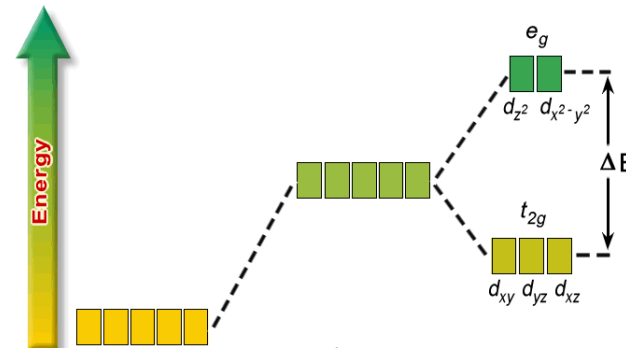
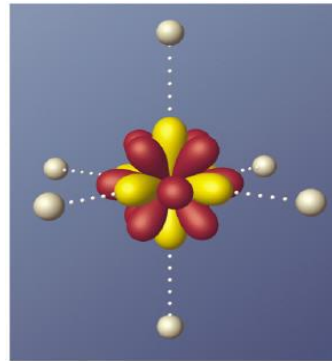
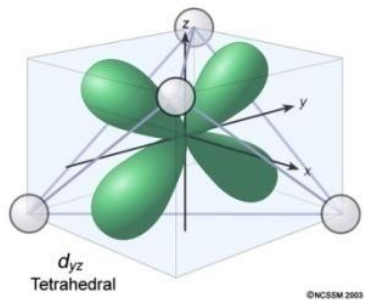
d^2sp^3 hybrid

$[\text{Co}(\text{NH}_3)_6]^{3+}$
(inner orbital or low spin complex)

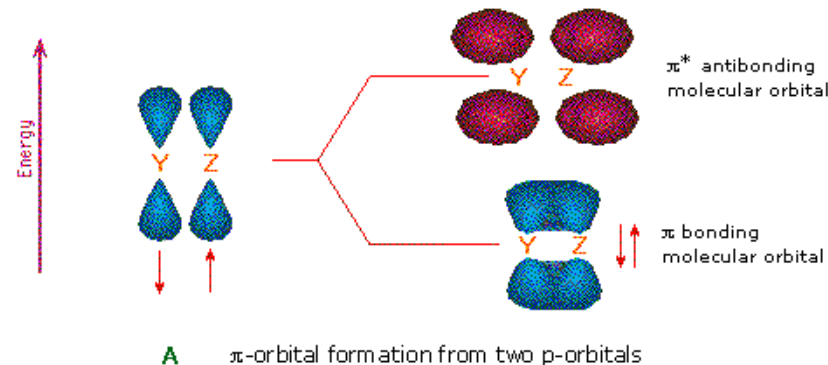


Six pairs of electrons from six NH_3 molecules

Crystal field theory

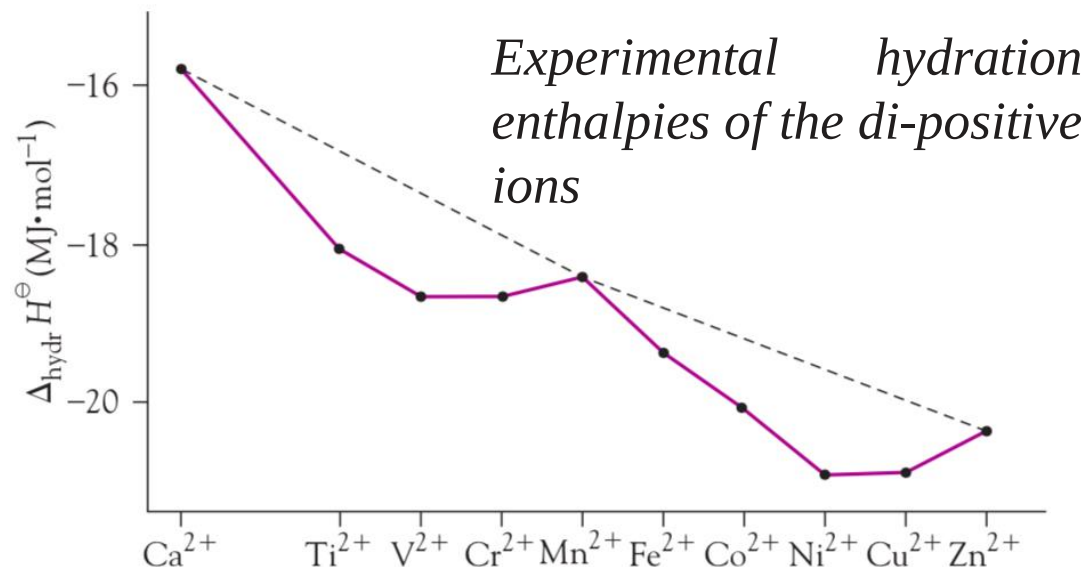


Molecular orbital Theory

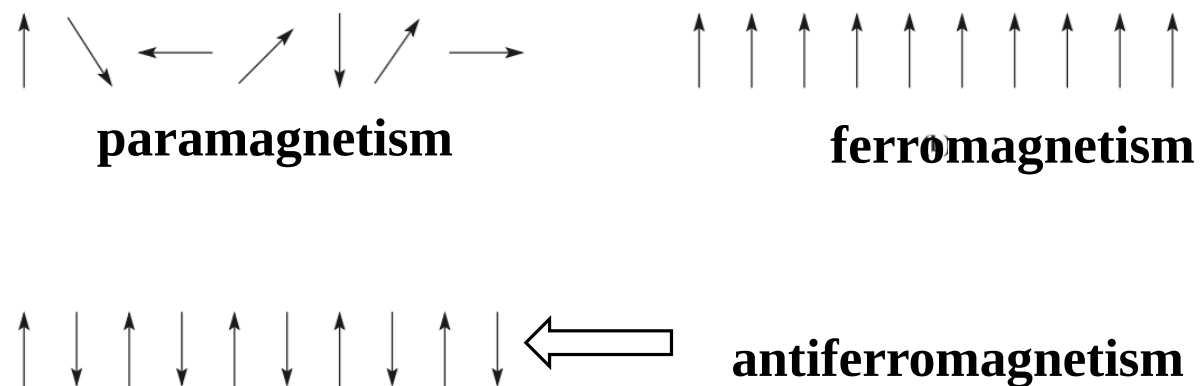


Importance of understanding bonding of transition metal complexes

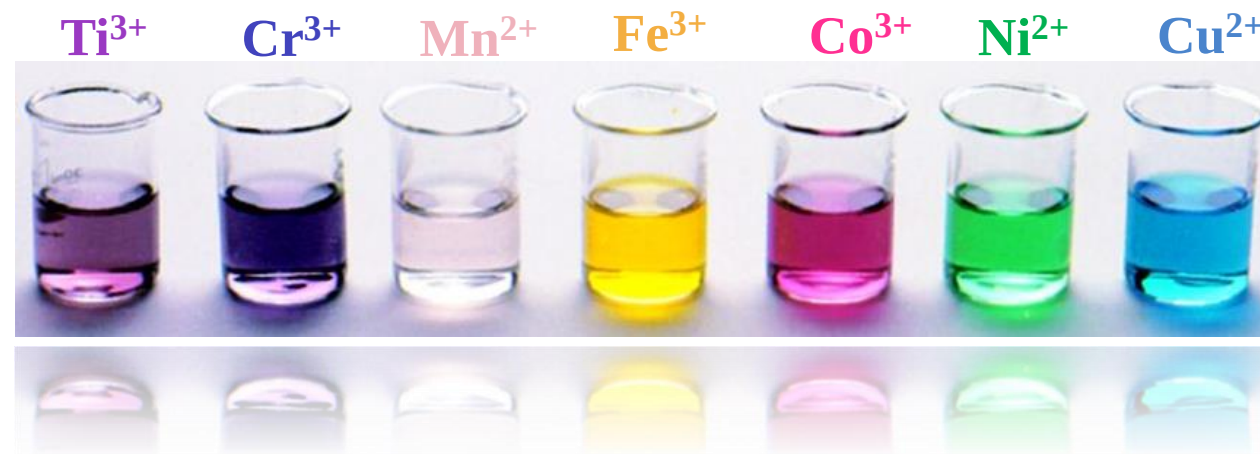
Physical properties of metal complexes



Magnetic properties of metal complexes

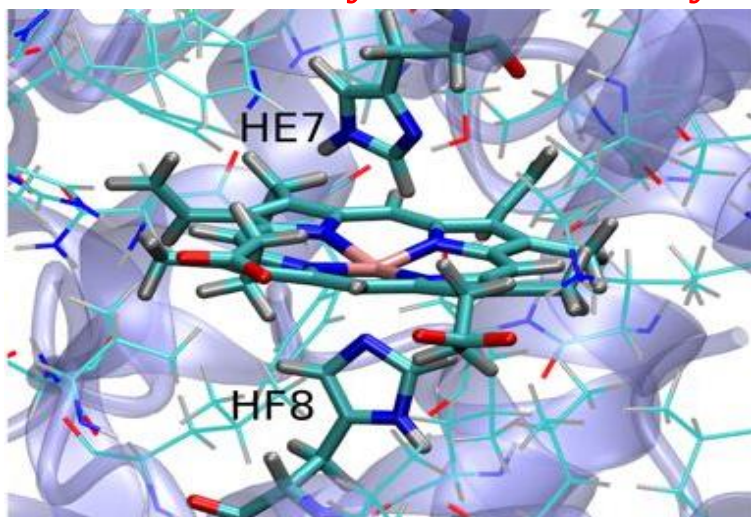


Color of transition metal complexes



Aqueous solution of different divalent/trivalent first-row transition metal ions

Structure and reactivity of metalloenzymes

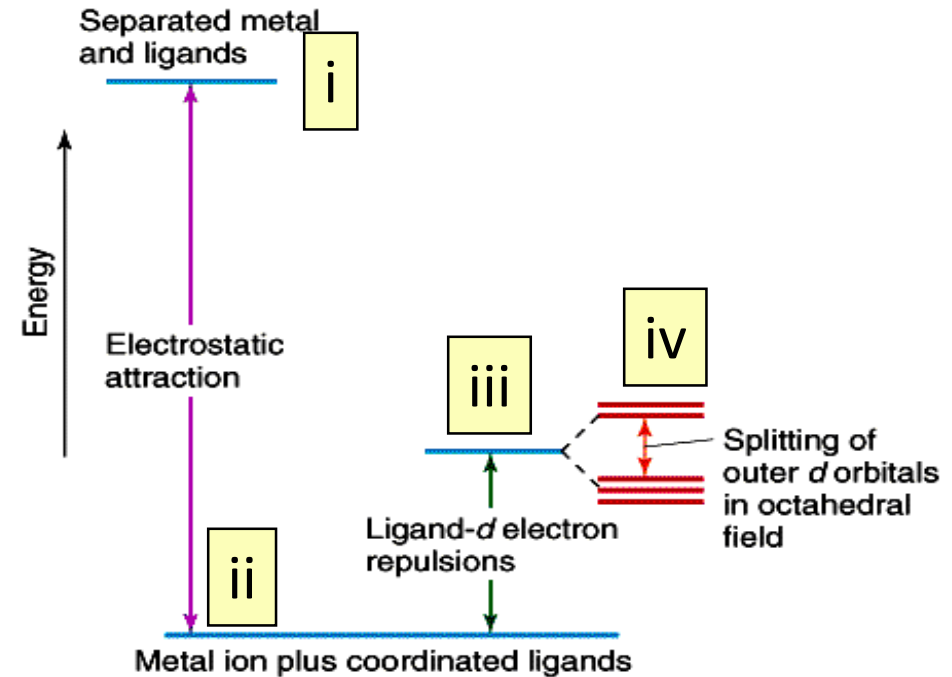


Energetics involved in the CFT model

Basic Assumptions of Crystal Field theory (concept borrowed from solid state physics)

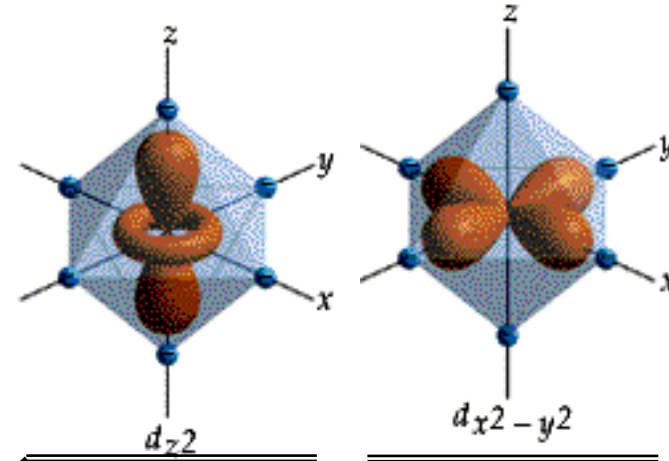
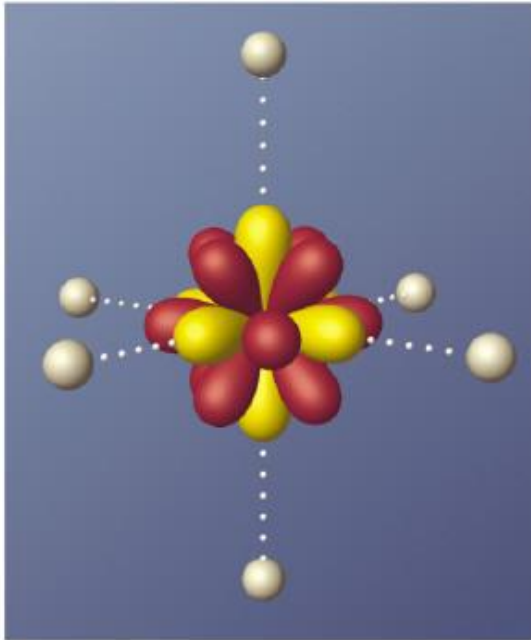
- The ligands and the metal are considered as point charges
- The attraction between the metal and the ligands is purely electrostatic
- Properties of the metal complexes are explained based on changes happening to the *d orbitals of the metal only*

- Separated metal and ligands at high energy
- Electrostatic attraction between metal and ligands leading to stabilization (**heart of CFT**)
- Destabilization due to ligand electrons- metal d electrons repulsion in a **spherical** field
- Further splitting of the 5 d orbitals to 2 sets due to an **octahedral** field.

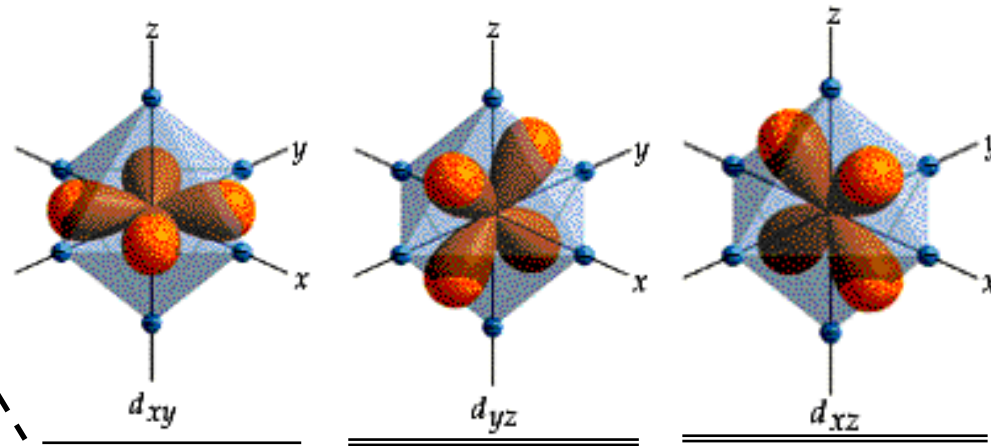


Orbitals affected when ligands approach a metal in an octahedral arrangement

Electrons in d_{z^2} and $d_{x^2-y^2}$ orbitals are concentrated close to the ligands, along the axes, resulting strong repulsions.

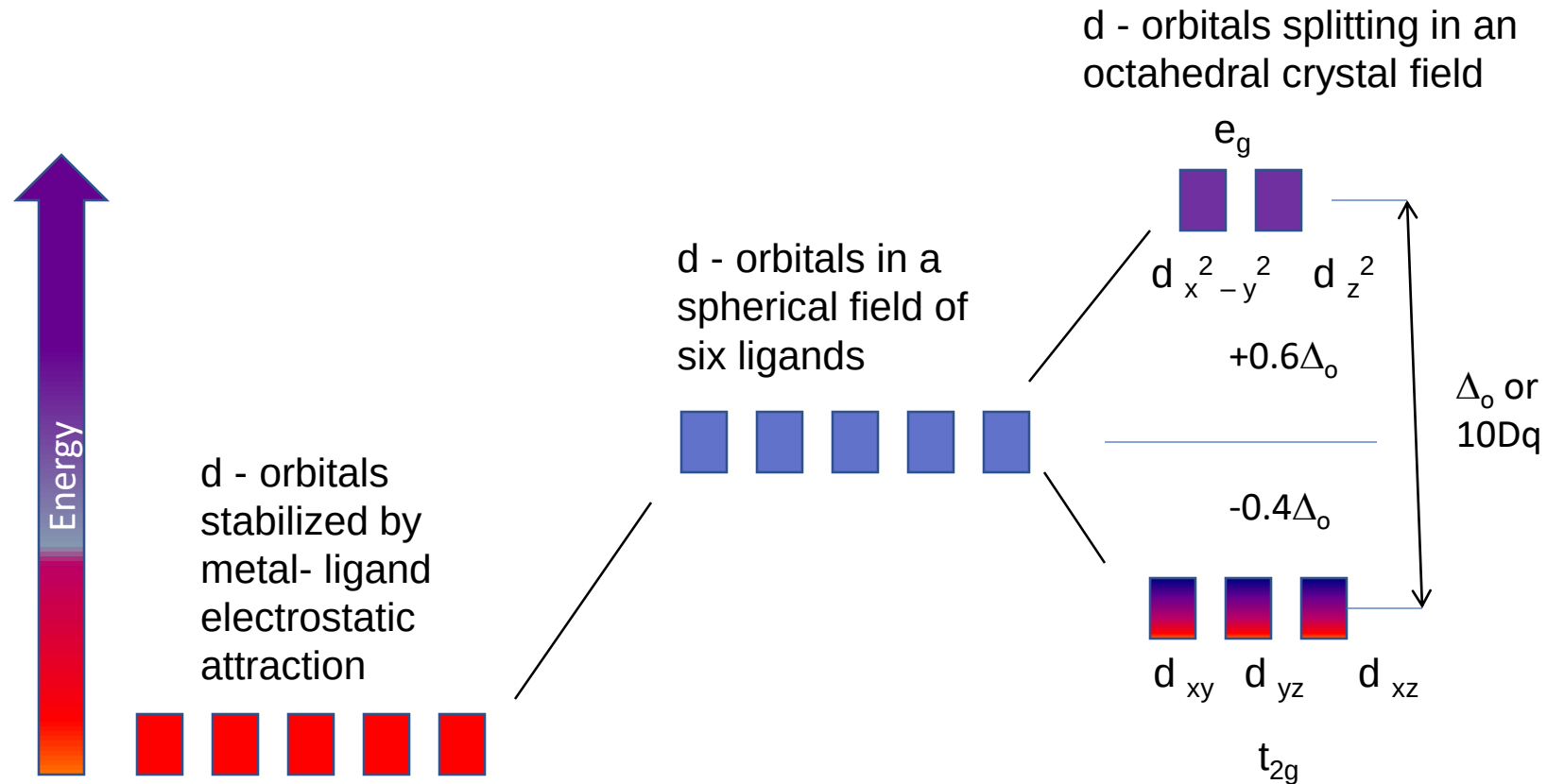


d-orbitals having lobes pointing directly at x, y and z axis are repelled most by electron-electron repulsion



d-orbitals not pointing directly at x, y and z axis are stabilized to maintain the overall energy same

Splitting of the d-orbitals in an octahedral field



t_{2g} : triply degenerate set of orbitals :

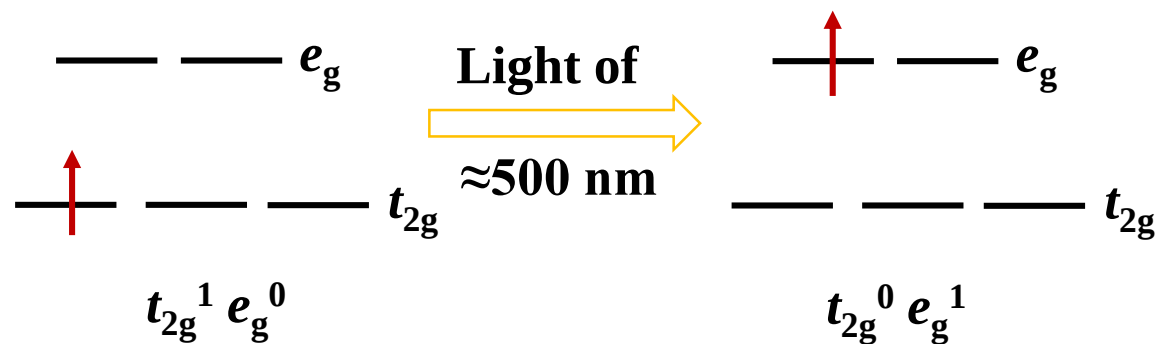
E_g : doubly degenerate set of orbitals

G = gerade: symmetric with respect to the centre of inversion

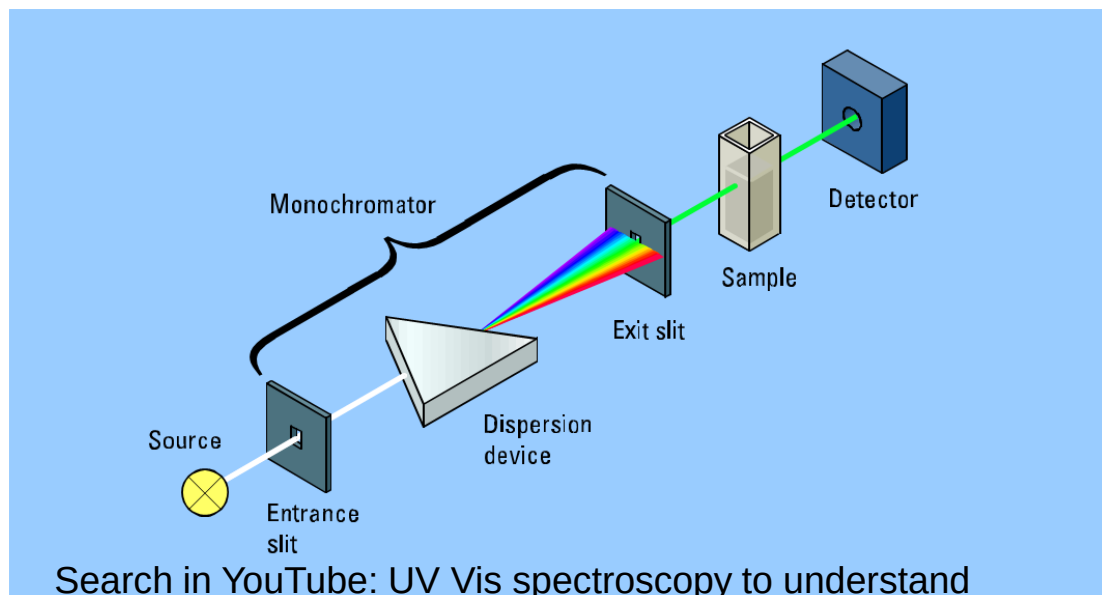
U = ungerade: antisymmetric w. r. t the centre of inversion

Barycenter: A point between objects where they balance each other

Δ_O is an experimental quantity and can be evaluated from electronic absorption spectroscopic data

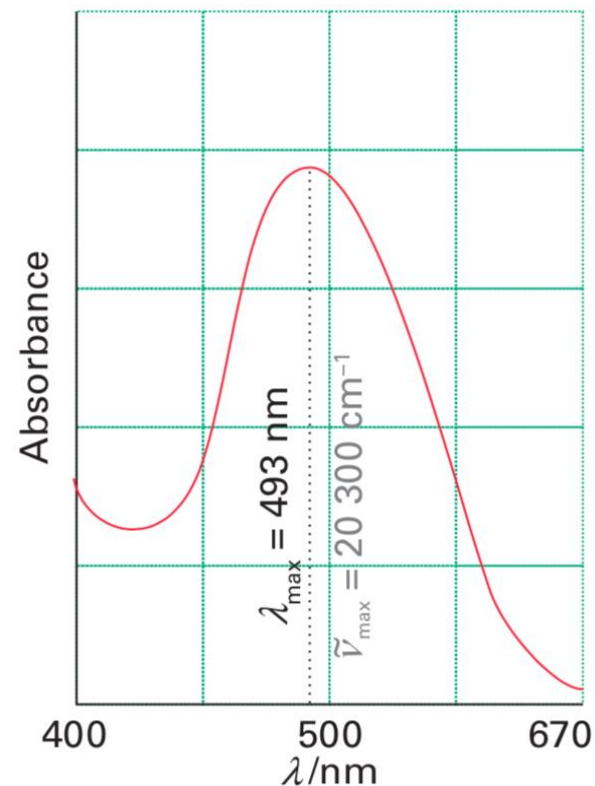


- Corresponding to an energy change of 243 kJ mol^{-1} corresponds to Δ_O ($1 \text{ cm}^{-1} = 11.96 \times 10^{-3} \text{ kJ/mole}$)



Search in YouTube: UV Vis spectroscopy to understand more about it.

The optical absorption spectrum of $[\text{Ti}(\text{OH}_2)_6]^{3+}$.

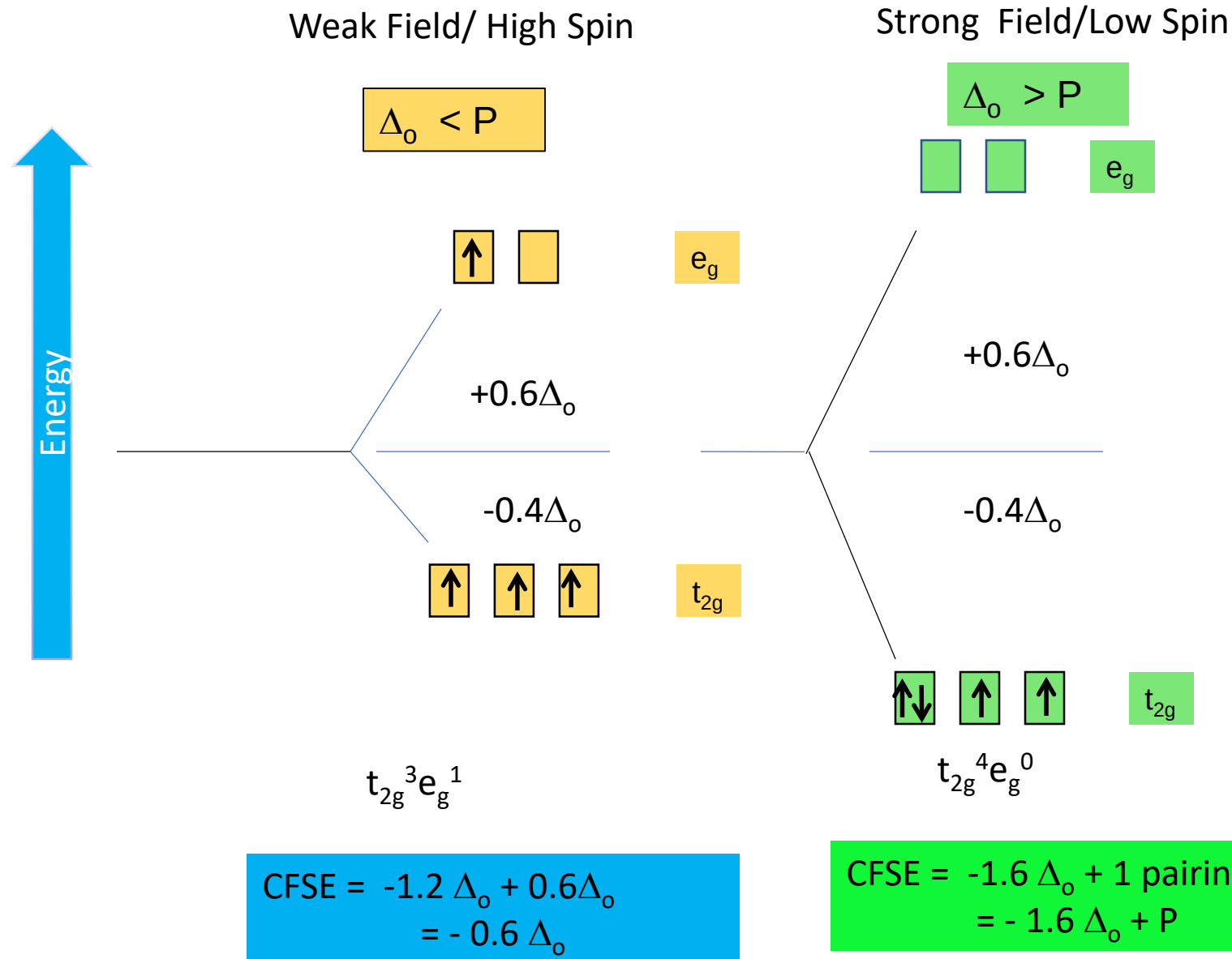


$$[\text{Ti}(\text{H}_2\text{O})_6]^{3+} \quad \Delta_O = 20,300 \text{ cm}^{-1} = 243 \text{ kJ/mol}$$

Since an electron in the t_{2g} set is stabilized by $-0.4 \Delta_O$
 $243 \times -0.4 = -97 \text{ kJ/mol}$

The complex is stabilized to the extent of 97 kJ/mol compared to a hypothetical spherical field due to the splitting of the d orbitals; This extra stabilization of the complex is called **crystal field stabilization energy (CFSE)**

Crystal Field Stabilization Energy : The d⁴ Case

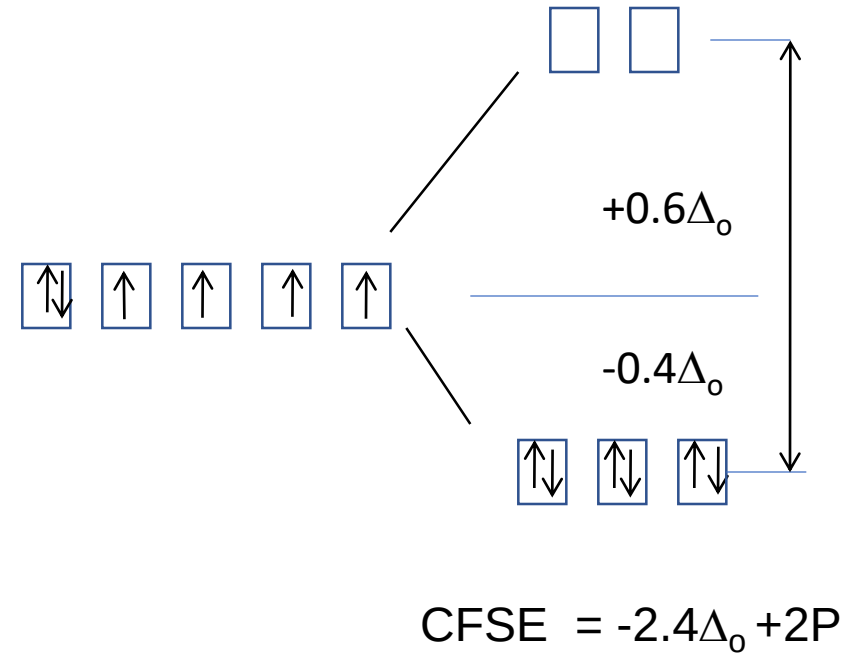
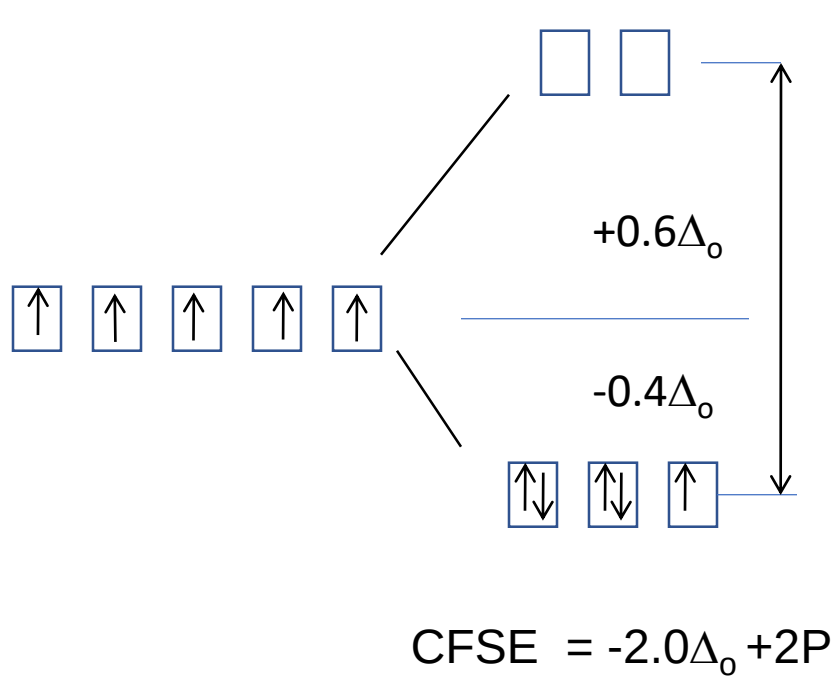


Hund's rule states that

Every orbital in a sublevel is singly occupied before any orbital is doubly occupied

The pairing energy is **not** an experimentally obtained value like Δ_o and is **same** for a metal ion irrespective of the ligands. (a typical value of P is about $15,000 \text{ cm}^{-1}$)

The d^5 and d^6 low spin cases and the correct way to determine pairing energy contribution

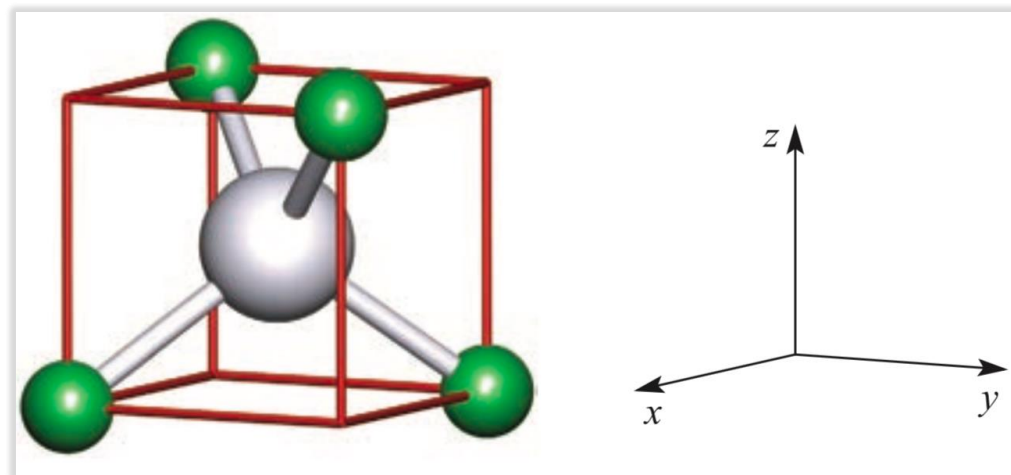


CFSE = $-2.4\Delta_o + 3P$

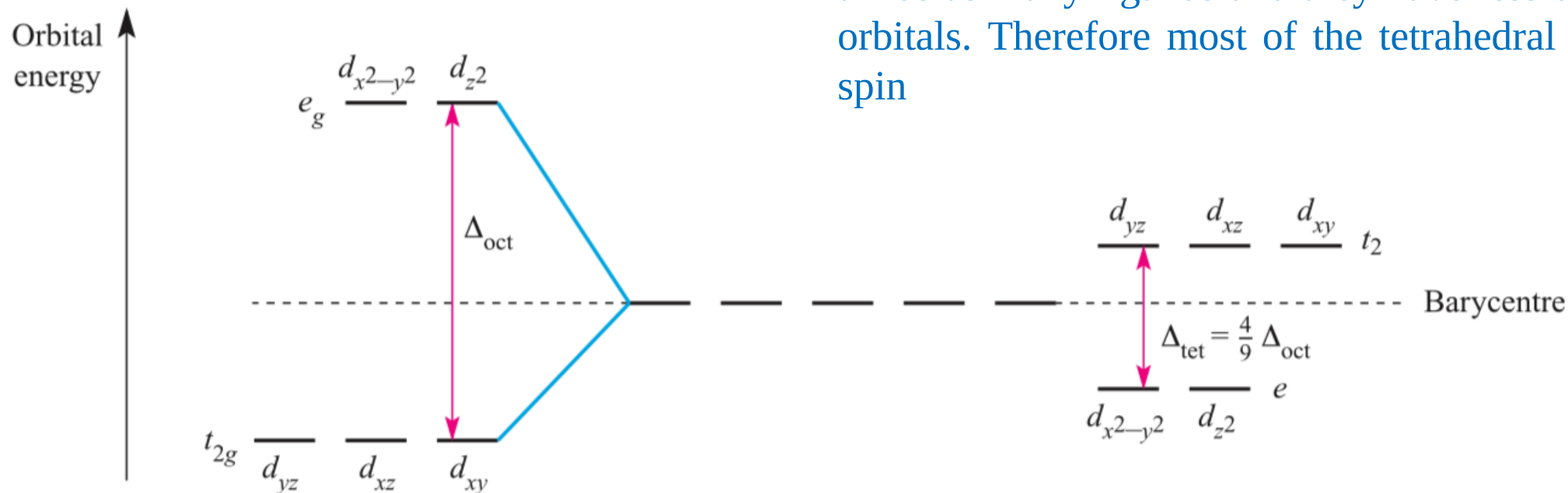


Always compare the extent of pairing with the situation before the splitting of five d orbitals occurred

Tetrahedral crystal field

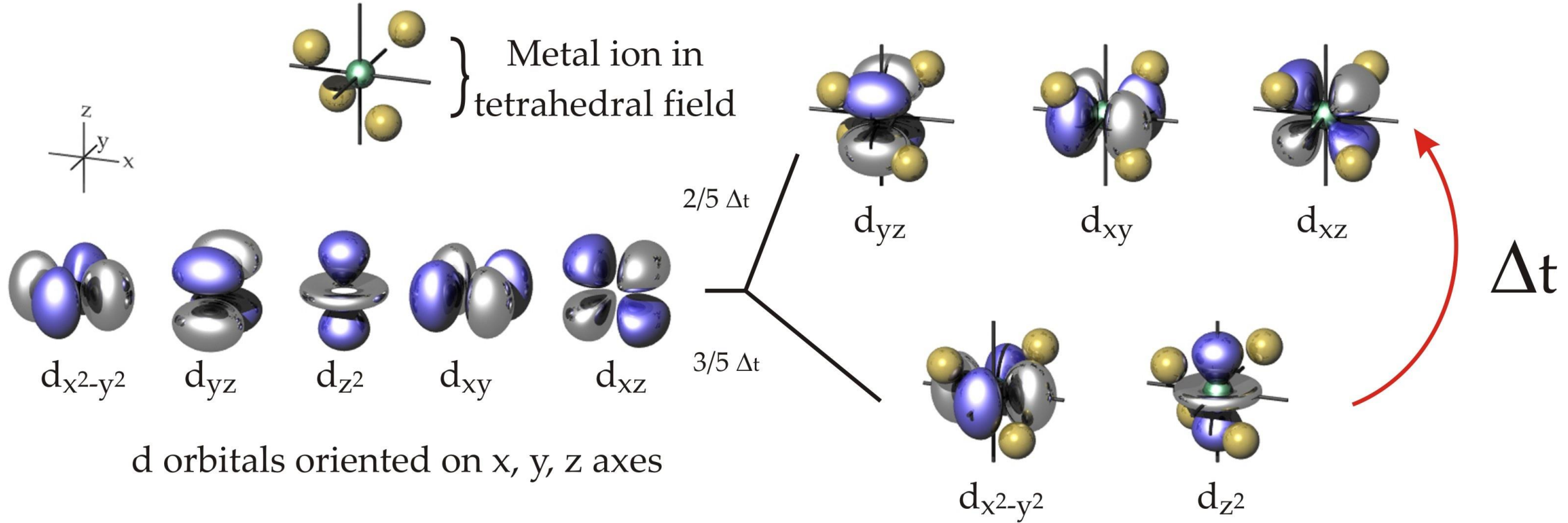


- In a tetrahedral complex, none of the metal d orbitals points exactly at the ligands, but the d_{xy} , d_{yz} and d_{xz} orbitals come nearer to doing so than the d_{z^2} and $d_{x^2-y^2}$ orbitals.
- The splitting of the d orbitals is therefore inverted compared with that for a regular octahedral structure, and the energy difference (Δ_t) is smaller.
- The crystal field splitting in a tetrahedral field is intrinsically smaller than that in octahedral field because there are only two-thirds as many ligands and they have less direct effect on the d orbitals. Therefore most of the tetrahedral complexes are high spin



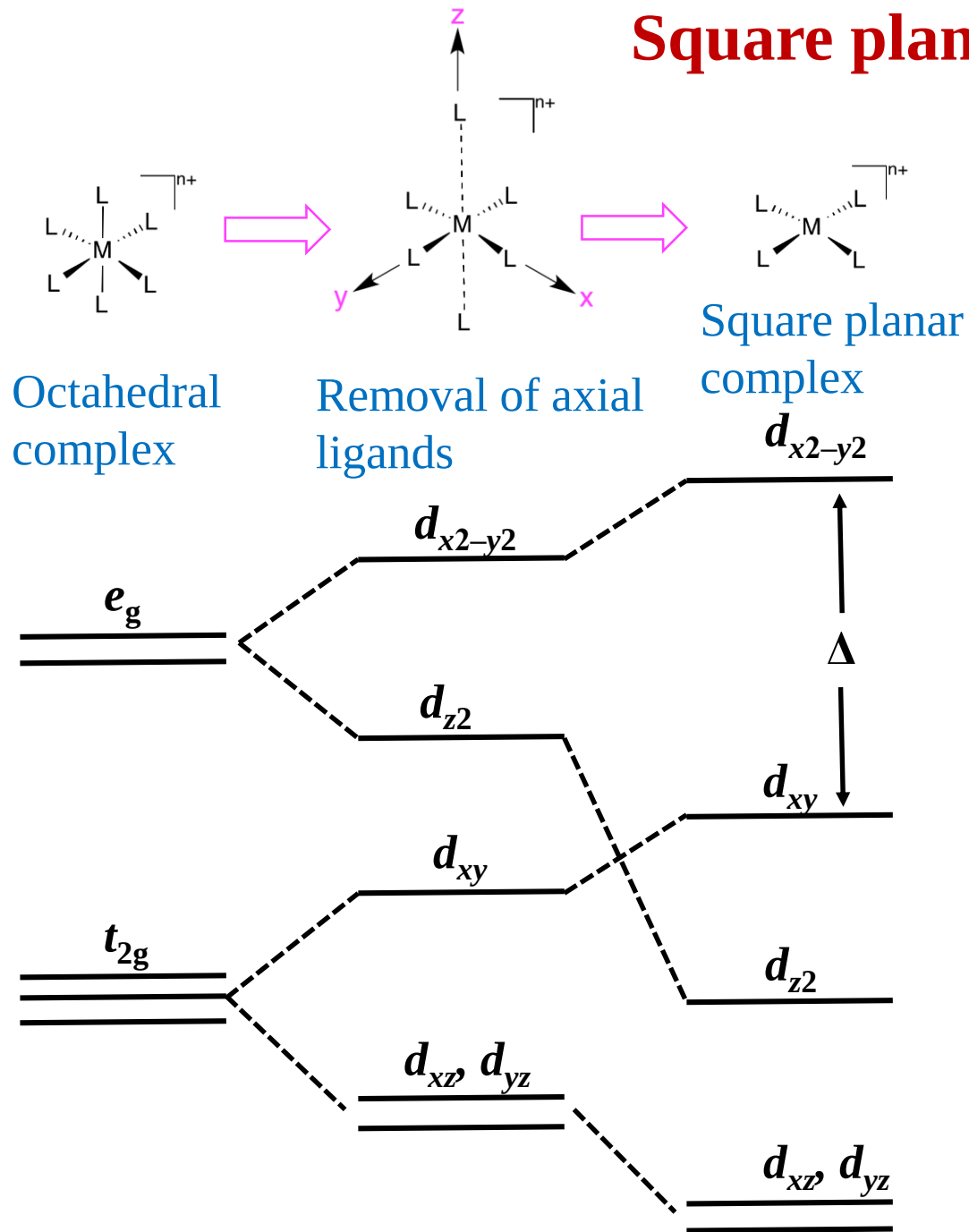
$$\Delta_t = \frac{4}{9} \Delta_o \approx \frac{1}{2} \Delta_o$$

Position of the d-orbitals in tetrahedral crystal field



d Orbital Splitting in a Tetrahedral Field

Square planar crystal field



➤ Square planar is related to octahedral splitting: Removing the axial ligands of an octahedron to infinity results in a square plane

➤ It is observed that $\Delta_{sp} > \Delta_o$ for complexes of the same metal and ligands with the same M–L bond lengths.

Second and third row metal d^8 complexes (e.g. Pt(II), Pd(II), Rh(I), Ir(I)) are invariably square planar.

$[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{PdCl}_4]^{2-}$, $[\text{Pt}(\text{NH}_3)_4]^{2+}$, $[\text{PtCl}_4]^{2-}$, $[\text{AuCl}_4]^{2-}$